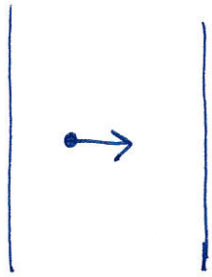
P1



a) $\frac{dp}{dt} = \frac{eV}{l} \Rightarrow p(t) = \frac{eVt}{l}$, $p = \frac{mv}{\sqrt{1-\frac{v^2}{c^2}}}$

$$\Rightarrow \frac{v^2}{1-\frac{v^2}{c^2}} = \left(\frac{eVt}{ml}\right)^2, \quad v^2\left[1+\left(\frac{eVt}{mcl}\right)^2\right] = \left(\frac{eVt}{ml}\right)^2$$

$$v(t) = \frac{eVt/ml}{\sqrt{1+\left(\frac{eVt}{mcl}\right)^2}} \Rightarrow \frac{v(t)}{c} = \frac{at}{\sqrt{1+(at)^2}}, \quad a = \frac{eV}{mlc}$$

b) Time of flight T: $l = \int_0^T v(t)\,dt = c\int_0^T \frac{at}{\sqrt{1+a^2t^2}}\,dt =$

$$(x=at) \quad = \frac{c}{a}\int_0^{aT} \frac{x}{\sqrt{1+x^2}}\,dx = \frac{c}{a}\left(1+x^2\right)^{1/2}\Big|_0^{aT} = \frac{c}{a}\left(\sqrt{1+a^2T^2}-1\right)$$

$$\cancel{1+a^2T^2 = \left(\frac{al}{c}\right)^2 \Rightarrow T = \sqrt{\left(\frac{l}{c}\right)^2 - \left(\frac{mlc}{eV}\right)^2} = \frac{l}{c}\sqrt{1-\left(\frac{mc^2}{eV}\right)^2}}$$

$$\sqrt{1+a^2T^2} = 1+\frac{al}{c}, \quad 1+a^2T^2 = 1+\frac{2al}{c}+\left(\frac{al}{c}\right)^2$$

$$T^2 = \frac{2l}{ac}+\left(\frac{l}{c}\right)^2 = \frac{l^2}{c^2}\left(1+\frac{2c}{al}\right)$$

$$T = \frac{l}{c}\sqrt{1+\frac{2c}{al}} = \frac{l}{c}\sqrt{1+\frac{2mc^2}{eV}} = \sqrt{\frac{2ml^2}{eV}+\frac{l^2}{c^2}}$$

$$\frac{v(T)}{c} = \frac{aT}{1+\frac{al}{c}} = \frac{\frac{al}{c}\sqrt{1+\frac{2mc^2}{eV}}}{1+\frac{al}{c}} = \frac{\frac{eV}{mc^2}\sqrt{1+2\frac{mc^2}{eV}}}{1+\frac{eV}{mc^2}} = \frac{\sqrt{\frac{2eV}{mc^2}+\left(\frac{eV}{mc^2}\right)^2}}{1+eV/mc^2}$$

Proper time $t' = \int_0^T dt \sqrt{1-\frac{v^2}{c^2}} = \int_0^T dt \sqrt{1-\frac{a^2t^2}{1+a^2t^2}} = \int_0^T \frac{dt}{\sqrt{1+a^2t^2}} =$

$$= \frac{1}{a}\int_0^{aT} \frac{du}{\sqrt{1+u^2}} = \frac{1}{a}\int_0 \frac{ch\varphi\, d\varphi}{\sqrt{1+sh^2\varphi}} = \frac{\varphi}{a}\Big|_{\cancel{0}} = \frac{1}{a} sh^{-1}(aT)$$

$$u = sh\varphi = \frac{e^{\varphi}-e^{-\varphi}}{2} \Rightarrow e^{\varphi} - \frac{1}{e^{\varphi}} = 2u$$

In more details:

$$(e^{\varphi})^2 - 2ue^{\varphi} - 1 = 0$$

$$e^{\varphi} = u \pm \sqrt{u^2+1} = u + \sqrt{1+u^2} \quad \left(e^{\varphi} > 0 \text{ always}\right)$$

$$\Rightarrow \varphi = \ln\left(u+\sqrt{1+u^2}\right)$$

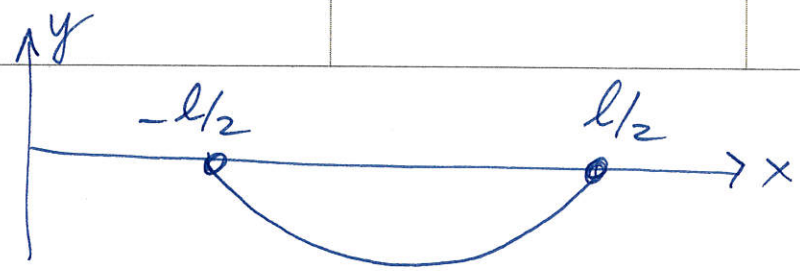
$$\Rightarrow t' = \frac{1}{a}\ln\left(aT+\sqrt{1+a^2T^2}\right) = \frac{1}{a}\ln\left(\sqrt{\frac{2al}{c}+\left(\frac{al}{c}\right)^2}+\frac{al}{c}+1\right)$$

$$\frac{al}{c} = \frac{eV}{mc^2}$$

$$t' = \frac{mcl}{eV}\ln\left(1+\frac{eV}{mc^2}+\sqrt{2\frac{eV}{mc^2}+\left(\frac{eV}{mc^2}\right)^2}\right) \longrightarrow$$

$$\longrightarrow \frac{mcl}{eV}\begin{cases}\sqrt{\frac{2eV}{mc^2}}, & \frac{eV}{mc^2} \ll 1 \quad \text{(c cancels in non-rel. limit)} \\ \ln\left(\frac{2eV}{mc^2}\right), & \frac{eV}{mc^2} \gg 1\end{cases}$$

P2



$$\text{Area} = \int_{-\ell/2}^{\ell/2} dx\; y(x) \quad , \text{ Rope length } L = \int ds = \int_{-\ell/2}^{\ell/2} dx \sqrt{1+(y')^2}$$

$\Rightarrow$ need to maximize functional (λ = Lagrange multiplier)

$$S = \int_{-\ell/2}^{\ell/2} dx \left(y(x) + \lambda \sqrt{1+(y'(x))^2} \right) = \int dx\, \mathcal{L}$$

choosen so that length = L.

EL eqn: $$\frac{\partial}{\partial y} \mathcal{L} - \frac{d}{dx} \frac{\partial \mathcal{L}}{\partial y'} = 0$$

$$1 - \lambda \frac{d}{dx} \frac{y'}{\sqrt{1+y'^2}} = 0$$

$$\Rightarrow \frac{y'}{\sqrt{1+y'^2}} = \frac{x-x_0}{\lambda} \quad (x_0 = \text{constant of integration})$$

$$\frac{y'^2}{1+y'^2} = \left(\frac{x-x_0}{\lambda}\right)^2 \Rightarrow y'^2 = \frac{\left(\frac{x-x_0}{\lambda}\right)^2}{1-\left(\frac{x-x_0}{\lambda}\right)^2} = \frac{(x-x_0)^2}{\lambda^2-(x-x_0)^2}$$

$$\frac{dy}{dx} = \pm \frac{x-x_0}{\sqrt{\lambda^2-(x-x_0)^2}} \quad (\text{choose } (-) \text{ sign on phys. grounds})$$

$$y-y_0 = -\int dx \frac{x-x_0}{\sqrt{\lambda^2-(x-x_0)^2}} = +\sqrt{\lambda^2-(x-x_0)^2}$$

$$\Rightarrow \quad (y(x)-y_0)^2 + (x-x_0)^2 = \lambda^2 \quad \leftarrow \text{circle of radius } \lambda \text{ centered at } (x_0, y_0)$$

Boundary conditions: $y(x = \pm \ell/2) = 0$

$$\Rightarrow \begin{cases} y_0^2 + \left(\frac{\ell}{2} - x_0\right)^2 = \lambda^2 \\ y_0^2 + \left(\frac{\ell}{2} + x_0\right)^2 = \lambda^2 \end{cases} \Rightarrow x_0 = 0.$$

Length $L = \int dx \sqrt{1 + \left(\frac{x}{\sqrt{\lambda^2 - x^2}}\right)^2} = \lambda \int_{-\ell/2}^{\ell/2} \frac{dx}{\sqrt{\lambda^2 - x^2}}$

$$\frac{dy}{dx} = -\frac{x}{\sqrt{\lambda^2 - x^2}} \quad (x_0 = 0)$$

let $x = \lambda \sin\varphi \Rightarrow L = \lambda^2 \int \frac{\cos\varphi \, d\varphi}{\lambda \sqrt{1 - \sin^2\varphi}} = \lambda\varphi = \lambda \sin^{-1}\left(\frac{x}{\lambda}\right)\Big|_{-\ell/2}^{\ell/2}$

$$\Rightarrow L = 2\lambda \sin^{-1}\left(\frac{\ell}{2\lambda}\right)$$ is implicit equation for $\lambda(L)$.

P3 $\vec{F} = -\frac{a}{x^3}\hat{x}$, $\vec{F} = -\frac{\partial U}{\partial x}\hat{x} \Rightarrow U(x) = \frac{-a}{2x^2}$

b) Energy $E = \frac{1}{2}mv^2 - \frac{a}{2x^2} = -\frac{a}{2x_0}$

$$\frac{1}{2}mv^2 = \frac{a}{2}\left(\frac{1}{x^2} - \frac{1}{x_0^2}\right) , \quad \frac{dx}{dt} = -\sqrt{\frac{a}{m}}\sqrt{\frac{1}{x^2} - \frac{1}{x_0^2}}$$

$$\int_{x_0}^{0} \frac{dx}{\sqrt{\frac{1}{x^2} - \frac{1}{x_0^2}}} = -\sqrt{\frac{a}{m}}\, t$$

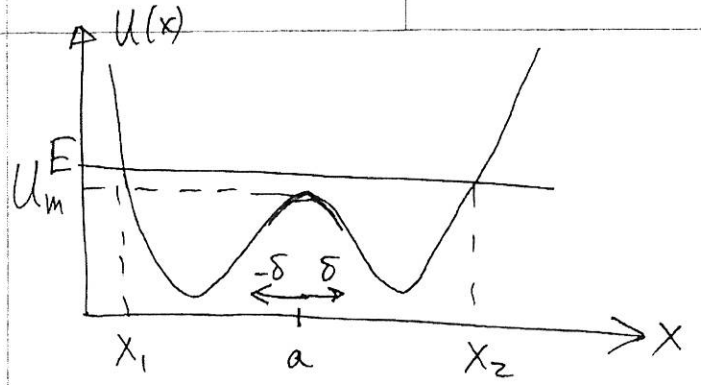
let $x = x_0 u \Rightarrow$ $$\int_0^1 \frac{x_0\, du}{\frac{1}{x_0}\sqrt{\frac{1}{u^2} - 1}} = \sqrt{\frac{a}{m}}\, t$$

$$\sqrt{\frac{a}{m}}\, t = x_0^2 \int_0^1 \frac{u\, du}{\sqrt{1-u^2}} = x_0^2$$

$$t = \sqrt{\frac{m}{a}}\, x_0^2 .$$

a) $L = \frac{1}{2}m\dot{x}^2 + \frac{a}{2x^2}$.

P4



$E - U_m \ll U_m$

$U(x) \approx U_m - \frac{1}{2} k (x-a)^2$

Particle's speed is smallest when $x = a \Rightarrow$ it spends most time there!

$$T = 2\sqrt{\frac{m}{2}} \int_{x_1}^{x_2} \frac{dx}{\sqrt{E - U(x)}} = T_1 + T_2$$

where $T_1 = \sqrt{2m} \int_{a-\delta}^{a+\delta} \frac{dx}{\sqrt{E - U(x)}}$ is the time of motion through $(a-\delta, a+\delta)$ interval,

T_2 is the rest of the integral.

Near $x = a$ $\quad E - U(x) = E - U_m + \frac{1}{2} k (x-a)^2$

$$T_1 = \sqrt{2m} \int_{a-\delta}^{a+\delta} \frac{dx}{\left(\frac{1}{2}k\right)^{1/2} \sqrt{\frac{2(E-U_m)}{k} + (x-a)^2}} = (x = a + y)$$

$$= \sqrt{\frac{4m}{k}} \int_{-\delta}^{\delta} \frac{dy}{\sqrt{\underbrace{\frac{2(E-U_m)}{k}}_{\alpha^2} + y^2}} = 4\sqrt{\frac{m}{k}} \ln\left(y + \sqrt{\alpha^2 + y^2}\right)\Big|_0^{\delta}$$

$$= 4\sqrt{\frac{m}{k}} \ln\left(\frac{\delta + \sqrt{\alpha^2 + \delta^2}}{\alpha}\right) \to 4\sqrt{\frac{m}{k}} \ln \frac{2\delta}{\sqrt{\frac{2}{k}(E - U_m)}}$$

The integral diverges logarithmically as $E \to U_m$

$\Rightarrow T_1 \gg T_2$ then, and $T \approx T_1$.